

to transmit to aeroplanes flying over water is making airline operators take a second look at satellites. Although a satellite-based system is more expensive, the returns will also be higher. Operators can offer better speeds and more services, to more customers. They can offer value-added services including video streaming, online shopping, hotel booking and local destination coupons.

Southwest, an airline industry leader in the US, recently started rolling out a satellite service by Row 44—a California-based company. With an introductory pricing of \$5 each way, the broadband service is now available on 180 flights. AirTran, currently a Gogo customer, will also move to satellites next year. United Airlines is working with Panasonic Avionics, JetBlue subsidiary LiveTV and ViaSat.

Technology on the move

Currently, there is a lot of activity on the satellite technology front. Russia recently launched ViaSat—its mega

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Ka vs Ku			
Ka and Ku are the main frequency bands used for satellite-based in-flight broadband. Here is a quick comparison of the two:			
Band	Frequency	Advantages	Disadvantages
Ku	12-18 GHz	- Time-tested for aviation applications - Many satellites available - Not greatly affected by weather conditions	- Lower frequency - Lower speeds - Due to applications like television, the available spectrum is low - Large, expensive antennae required on flights
Ka	26.5-40 GHz	- Higher frequency - Higher speeds - A lot of free spectrum - Low-cost, small antennae on flights	- Still new to airborne applications - Affected by bad weather - Very few Ka-band satellites are currently available

satellite directed at broadband communications. A recent deal between the British satellite telecommunications company Inmarsat and US-based Honeywell is also expected to give in-flight broadband a big boost.

Inmarsat plans to launch three satellites in the near future. The first satellite that is expected to be launched in 2013, will provide global coverage in the Ka frequency band. This band will provide much faster data rates than Ku- and L-band frequencies, which were used by Boeing in its 2003 experiment.

Honeywell is all set to exploit the capability of these satellites with its Global Xpress project, which aims to make in-flight Wi-Fi fast, cheap, reliable and available anywhere. The company will be developing the required network hardware such as the antennae to send and receive satellite signals.

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Well, with all this activity in the satellite-based in-flight broadband space, it is not surprising that Gogo is exploring such an option! On May 18, 2012, Gogo—which powers the Internet on 85 per cent of American flights—announced that it will offer Ku-band satellite technology by the year-end as an interim move to support long-haul flights. By 2014, it will also provide airlines with Ka-band satellite technology to provide additional capacity to supplement its ATG and ATG-4 implementations.

Way to go, satellites! It is a pity Boeing's Internet service didn't get much oomph in 2003. Otherwise, we would already have by now what we wanted. In the next issue, we shall go into more details about why Boeing's attempt failed and why similar solutions appear to be successful now. We shall also compare the two in-flight Internet options—cellular and satellite—and see why airlines are moving from the former to the latter. ●

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